

PROBLEM

The problem: the EnKF's uncertainty had collapsed

Ensemble spread ($\hat{\sigma}$)

0.0025

Actual error (RMSE)

0.24

$\hat{\sigma}$ is the spread of the 100 members. It was about 1% of the real error: the filter was a hundred times too confident.

So the robots were ignored. The Kalman update moved the estimate only 0.2% of the way towards the observations.

The cause: only 1% of the process noise was added

```
X_f += rng.normal(0, 0.01 * proc_noise_vec)
```

original forecast step

The forecast model pulls the members together every step; the spread settles at the size of the noise that is added back. At $0.01 \times$ that is 1% of the error.

Noise multiplier	Spread / RMSE	CRPS ↓
0.01 (original)	0.013	0.122
1 (full strength)	0.956	0.097

Changing only this constant already fixes the size of the uncertainty. But with independent noise per cell, $\hat{\sigma}$ was nearly flat in space and could not show where the errors are.

The fix: add real forecast errors as the noise

1 Record the model's one-step errors

$$r = x(t+1) - f(x(t-4), \dots, x(t))$$

32 training days $\rightarrow$ 8,192 whole error maps, all four channels

2 Remove their average

$$r_k \leftarrow r_k - \bar{r}$$

keeps only the random part; the bias correction handles the systematic part

3 Each step, each member adds one map

drawn at random from the bank

A real error map is large on the walkways, zero at the walls, and density and velocity err together. The ensemble spread inherits that shape.

-17.6%

CRPS vs. Gaussian noise of the same size

The noise step, for each member i

$$1 \quad \varepsilon_i \sim \text{bank}$$

$$2 \quad \eta_i(t) = \rho \cdot \eta_i(t-1) + \sqrt{(1-\rho^2)} \cdot \varepsilon_i$$

$$3 \quad q_i = 1.5 \cdot b \odot \eta_i$$

$$4 \quad q_i \leftarrow q_i - \text{mean}_j q_j$$

$$5 \quad x_i \leftarrow f(x_i) - \text{bias} + q_i$$

ε_i	a random error map from the bank
η_i	member i 's noise, with memory over time
$\rho = 0.5$	share of last frame's noise that is kept
$\sqrt{(1-\rho^2)}$	keeps the noise size unchanged
q_i	the noise actually added to member i
1.5	overall size factor
$b \odot$	per-cell factor: 1 where a robot sees; on unseen cells 1, 1.25, 1.4, 0.93 for density, vx, vy, variance
mean_j	average over the 100 members, so the noise spreads them without shifting them
f, bias	forecast model; learned systematic error

What changed from the original filter

	Original	Final
Forecast model	original model, fed 1 frame	our retrained model, 5 frames
Process noise	independent per cell, $\times 0.01$	real error maps, measured size $\times 1.5$
		persist over time, more on unobserved cells
Kalman update	radius 7, inflation 1.02	same + cross-channel weights
Bias correction	yes	same

Teal: changed. Everything else is identical, so the results on the next slide come from these changes.

Result: original vs. final, scored identically

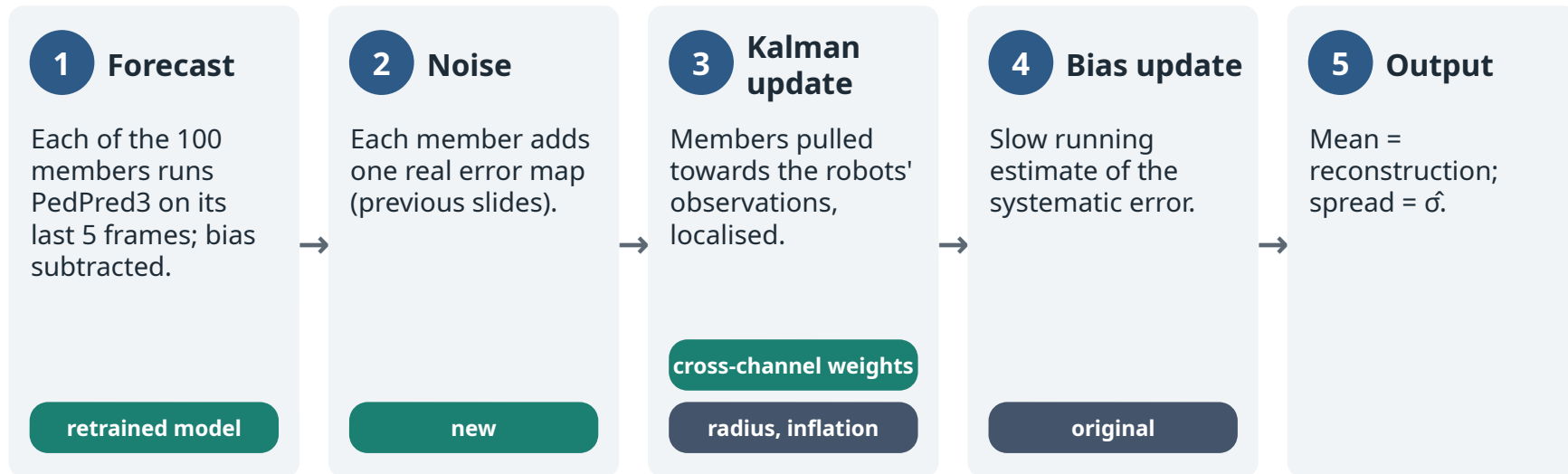
Unobserved cells, 4 channels	Original	Final	Ideal
RMSE (lower is better)	0.264	0.262	—
Spread / RMSE	0.010	1.008	1
CRPS skill (higher is better)	-0.252	0.113	> 0

Uncertainty: unusable → usable. Spread from 1% of the error to 1.01×.

Accuracy kept. RMSE slightly lower; velocity variance is still 13% worse than the original.

SUMMARY

The final filter, every frame



↻ step 5 feeds step 1 of the next frame (one second later)